

Updates from Silimate

DAC 2026: Open-Source EDA Birds-of-a-Feather Session
07/28/26

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Overview

Updates from Silimate

- About **Silimate** and how we rely on open-source EDA
- Results from our first **open-source bounty program**
- Updates on **Yosys** + **yosys-slang** (now called **sv-elab**)
- Updates on **OpenSTA**
- Three new **open-source EDA utilities** from Silimate
 - **Liberty2JSON**: Liberty file frontend for EDA tools
 - **PruneCCS**: stripping CCS and other info from Liberty files
 - **PruneFL**: SystemVerilog file list pruning for scalability

Silimate Products & Open-Source

What we build and ship

- **Preqorsor: PPA optimization**
 - PPA estimation and bottleneck analysis: 40-1000x faster using AI
 - RTL optimization in the loop
- **Silimate Debugger (SMDB)**
 - Root-cause analysis (RCA)
 - RTL testbench fixes in the loop
- **Silimate Copilot: Agentic harness for chip designers**



Open-Source Bounty Program

We ran our open-source bounty program for 6 months

- **Goals:**

- Outsource development of new features in open-source tools
- Outsource efforts to work with upstream maintainers to merge our contributions
- Source talent and increase contributor base for EDA tools

- **Results:**

- Massive success
- **28 contributions to Yosys, 2 contributions to GTKWave, 5 contributions to OpenSTA**
- Successfully hired several of the contributors

SV-Elab: SystemVerilog plugin for Yosys

The marriage between Yosys and Slang (formerly yosys-slang)

- **Key idea:** Slang elaborates hierarchy, resolves constants, types, and name references; sv-elab performs symbolic execution, emits word-level RTLIL
- Developed by Martin Povišer (povik), formerly YosysHQ, now at Precision Inno
- Since our 2025 presentation @ DAC BOF, **adoption has skyrocketed**
- Now integrated into mainline Yosys!

- **Supporters/contributors:**

ETH zürich



SILIMATE



YosysHQ



povik/**sv-elab**

SystemVerilog design elaborator into word-level netlist form



24
Contributors

57
Issues

11
Discussions

236
Stars

53
Forks



Updates on OpenSTA

Summaries of key contributions/updates

parallaxsw/
OpenSTA

Parallax
software

33 Contributors 8 Issues 16 Discussions 118 Stars 59 Forks

- **OpenSTA v3.0 release:** support for MMC, time borrowing, much more...
- **From Silimate:**
 - **Liberty / units / JSON:** parser and units hardening plus JSON path reporting — #279, #280, #285, #349, #354; also small build/test fixes #303, #305, #329, #366
 - **CI / Docker:** macOS + CentOS 7 Docker coverage landed in #473; Dockerfile mirror reliability in #460
 - **Bugfixes:** SWIG Seq double-free #470, slash ports #463, Liberty escaped/\$ bus names #464, non_seq cycle accounting #462
 - **Features open:** -filter_by_word #467, VCD begin/end #466, disable clock-gating check #465, max dynamic/leakage power #471, Liberty-backed properties #472

PruneFL

SystemVerilog file list pruning

Silimate/**pruneFL**

SystemVerilog file list pruner



2
Contributors

0
Issues

18
Stars

1
Fork



- **Problem:** massive file lists can slow down tool runtime significantly
- **Observation:** often, not all files are needed to elaborate
- **Solution:** prune all files that are not needed
- Uses **Slang** under the hood to elaborate, find dependencies, and clean up list of SystemVerilog files needed for compilation
- In production at **Tenstorrent**, resulting in massive speedups for EDA tool compilation times!

PruneCCS

Prune CCS (and other) data out of Liberty files

Silimate/**pruneccs**

Prune CCS data out of Liberty file for more compact files



0
Contributors

0
Issues

0
Stars

0
Forks



- **Problem:** Liberty files can be huge, making them a burden to store on disk or to load into memory
- **Observation:** Liberty files have tons of data, most of which may be useless for a particular application
- **Solution:** Prune unnecessary information out of libraries
- Designed for CCS data pruning, but easily extensible to other data in Liberty files (e.g., noise/variation data, so forth)

Liberty2JSON

Seamless Liberty frontend for the masses

Silimate/
liberty2json



Liberty to JSON converter

2
Contributors

0
Issues

3
Stars

0
Forks



- **Liberty: open-source standard for PDK cells/macros; developed before JSON!**
 - Format is very similar to JSON, but not quite the same
- We close that gap using the **battle-tested OpenSTA Liberty parser**
- Designed to allow seamless integration of Liberty into new EDA tools
- Additionally, have a **Yosys frontend** to import blackboxes
- **Deployed in production in Preqorsor (PPA tool)**
 - Works well with our Fortune 500 customers on latest commercial PDKs!

Thank you!

Silimate is hiring: <https://www.silimate.com>

- **Building the copilot for chip designers**
- **We love, use, and support open-source EDA (OpenSTA, Yosys, etc.)**
- **Well-funded and backed by Y Combinator, profitable in 2025, growing fast!**
- **Used and loved by companies like SiFive, Tenstorrent**
 - And many more (including Fortune enterprises) we can't disclose yet...
- **Come chat with me! Stop by our DAC booth #853**
 - Also, my email is akash@silimate.com

